



Solve with us 2.1 - Not Graded



This assignment will not be graded and is only for practice.

1. Key Points:

Calculating minors and cofactors of a matrix:

- If A is a square matrix, then the minor of the entry in the i -th row and j -th column (denoted by M_{ij}) is the determinant of the submatrix formed by deleting the i -th row and j -th column.
- The i -th j -th cofactor (denoted by C_{ij}) is defined to be $(-1)^{i+j} M_{ij}$.

Calculating the determinant:

- $\det(A) = \sum_{i=1}^n a_{ij} (-1)^{i+j} M_{ij} = \sum_{i=1}^n a_{ij} C_{ij}$ $\det(A) = \sum_{i=1}^n a_{ij} (-1)^{i+j} M_{ij} = \sum_{i=1}^n a_{ij} C_{ij}$
- Let us consider a 3×3 matrix A as follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Calculating M_{11} and C_{11} :

- Find out the sub-matrix by deleting the first row and the first column:

$$\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$$

- Calculating the determinant of the sub-matrix:

$$\det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} = a_{22} a_{33} - a_{32} a_{23}$$

- $M_{11} = a_{22} a_{33} - a_{32} a_{23}$
- $C_{11} = (-1)^{1+1} (a_{22} a_{33} - a_{32} a_{23}) = a_{22} a_{33} - a_{32} a_{23}$

Calculating M_{23} and C_{23} :

- Find out the sub-matrix by deleting the second row and the third column:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}$$

- Calculating the determinant of the sub-matrix:

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix} = a_{11} a_{32} - a_{31} a_{12}$$

- $M_{23} = a_{11} a_{32} - a_{31} a_{12}$
- $C_{23} = (-1)^{2+3} (a_{11} a_{32} - a_{31} a_{12}) = -a_{11} a_{32} + a_{31} a_{12}$

Calculating the determinant of A :



- Expanding with respect to the first row:

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

- Expanding with respect to the second row:

$$\det(A) = a_{21}(-1)^{2+1}M_{21} + a_{22}(-1)^{2+2}M_{22} + a_{23}(-1)^{2+3}M_{23}$$

$$\det(A) = a_{21}(-1)^{2+1}M_{21} + a_{22}(-1)^{2+2}M_{22} + a_{23}(-1)^{2+3}M_{23}$$

- Expanding with respect to the third row:

$$\det(A) = a_{31}(-1)^{3+1}M_{31} + a_{32}(-1)^{3+2}M_{32} + a_{33}(-1)^{3+3}M_{33}$$

$$\det(A) = a_{31}(-1)^{3+1}M_{31} + a_{32}(-1)^{3+2}M_{32} + a_{33}(-1)^{3+3}M_{33}$$

The determinant can also be calculated by expanding along columns.

- Expanding with respect to the first column:

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{21}(-1)^{2+1}M_{21} + a_{31}(-1)^{3+1}M_{31}$$

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{21}(-1)^{2+1}M_{21} + a_{31}(-1)^{3+1}M_{31}$$

Consider a square matrix $A = \begin{bmatrix} -12 & 0 \\ 2 & 0 \\ -10 & 1 \end{bmatrix}$. Find out M_{23} .

[Hint: Find out the sub-matrix by deleting the second row and the third column: $\begin{bmatrix} -12 & 0 \\ -10 & 1 \end{bmatrix}$]

Calculate the determinant of the sub-matrix.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Consider a square matrix $A = \begin{bmatrix} -12 & 0 \\ 2 & 0 \\ -10 & 1 \end{bmatrix}$. Find out C_{32} .

[Hint: Find out the sub-matrix by deleting the third row and the second column: $\begin{bmatrix} -12 & 0 \\ -10 & 1 \end{bmatrix}$]

Calculate the determinant of the sub-matrix to find M_{32} . To find the cofactor C_{32} the sign has to be taken care of as follows: $C_{32} = (-1)^{3+2}M_{32} = (-1)^{3+2}M_{32}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

Consider a square matrix $A = \begin{bmatrix} -12 & 0 \\ 2 & 0 \\ -10 & 1 \end{bmatrix}$. Find out the determinant of A .

[Hint: Expanding with respect to the first row:

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

Expanding with respect to any row will also give you the answer]

No, the answer is incorrect.**Score: 0****Accepted Answers:**

(Type: Numeric) -2

1 point

1 point

Consider the following square matrices:

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

$$A = \begin{bmatrix} 2013 & 2016 & 2019 \\ 2014 & 2017 & 2020 \\ 2015 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2013 & 2019 \\ 2017 & 2014 & 2020 \\ 2022 & 2015 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 2013 & 2019 \\ 4034 & 2014 & 2020 \\ 4044 & 2015 & 2021 \end{bmatrix}$$

Choose the set of correct options.

Hint:

- BB can be obtained by interchanging the first two rows.
- CC can be obtained by multiplying 2 to the first row of BB .

☐

$$\det(B) = \det(A) \det(B) = \det(A).$$

☐

$$\det(A) = -\det(B) \det(A) = -\det(B).$$

☐

$$\det(C) \neq -2\det(B) \det(C) = -2\det(B).$$

☐

$$\det(C) = -2\det(A) \det(C) = -2\det(A)$$

No, the answer is incorrect.**Score: 0****Feedback:**

- Sign of the determinant changes due to interchange of two rows.
- If a scalar c is multiplied with a row of a matrix, then the determinant of the new matrix will be c times the determinant of the earlier matrix.

Accepted Answers:

$$\det(A) = -\det(B) \det(A) = -\det(B).$$

$$\det(C) \neq -2\det(B)\det(C) = -2\det(B).$$

$$\det(C) = -2\det(A)\det(C) = -2\det(A)$$

2. Key Points:

Cramers' Rule: (This rule is applied to any system of linear equations with n equations and n variables. Here we recall the method for a system of linear equations with 3 equations and 3 variables.)

- Consider a system of linear equations as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

$$x_3 = \frac{b_1a_{21}x_1 + a_{22}x_2 + a_{23}x_3}{a_{31}x_1 + a_{32}x_2 + a_{33}x_3} = \frac{b_2a_{31}x_1 + a_{32}x_2 + a_{33}x_3}{b_3}$$

- Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

- Let A_{x_i} be the matrix obtained by replacing the i -th column of A (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b , for $i = 1, 2, 3$.

$$A_{x_1} = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}$$

$$A_{x_2} = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}$$

$$A_{x_3} = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$$

If $\det(A) \neq 0$, then the solutions to the above system are $x_i = \frac{\det A_{x_i}}{\det A}$

$x_1 = \frac{\det A_{x_1}}{\det A}$, $x_2 = \frac{\det A_{x_2}}{\det A}$, $x_3 = \frac{\det A_{x_3}}{\det A}$

Consider a system of linear equations
$$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$

Let matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by

replacing the i -th column of A (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b , for $i = 1, 2, 3$.

Use the above information to answer the following questions(5,6):

1 point

Choose the set of correct options

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$$A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} Ax_1 = \begin{bmatrix} 1-1011-10-11 \end{bmatrix}$$

☐

$$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} Ax_1 = \begin{bmatrix} 11101-11-11 \end{bmatrix}$$

☐

$$A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} Ax_2 = \begin{bmatrix} 1-1001-11-11 \end{bmatrix}$$

☐

$$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} Ax_3 = \begin{bmatrix} 1-1001-1111 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} Ax_1 = \begin{bmatrix} 11101-11-11 \end{bmatrix}$$

$$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} A x_3 = \begin{bmatrix} 1 & -1 & 0 & 0 & 1 & -1 & 1 & 1 & 1 \end{bmatrix}$$

1 point

Choose the set of correct options about the solutions to this system.

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$$x_1 = -2, x_2 = -2.$$

☐

$$x_2 = -2, x_3 = -2.$$

☐

$$x_3 = 3, x_3 = 3.$$

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x_1 = -2, x_2 = -2.$$

$$x_3 = 3, x_3 = 3.$$

3. Key Points:

Calculating the inverse of a matrix A:

- Find out the cofactor matrix CC , whose ij -th element is C_{ij} : the ij -th cofactor of AA .
- Adjoint of AA is transpose of CC .
- Calculate the determinant of AA and check whether it is non-zero or not.
- If determinant is non-zero then the inverse of AA exists and is given by

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) A^{-1} = \det(A)^{-1} \text{adj}(A).$$

1 point

Which of the following is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$?

[Hint: Recall the steps to calculate the cofactors, given in Key Points 1.]

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$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$$

☐

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$$

☐

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} \\ -1121-3011-2 \\ \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 3 & -1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} \\ 1-1-2-130-1-12 \\ \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} \\ -1121-3011-2 \\ \end{bmatrix}$$

1 point

Which of the following is the adjoint matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$?

[Hint: Find the transpose to the cofactor matrix.]

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$$\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \\ 111-1-3-1-202 \\ \end{bmatrix}$$

☐

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} \\ -1121-3011-2 \\ \end{bmatrix}$$

☐

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} \\ -1111-3120-2 \\ \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \\ 1-1-1-13-1-202 \\ \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} \\ -1111-3120-2 \end{bmatrix}$$

Find out the determinant of $\begin{bmatrix} 323 \\ 101 \\ 211 \end{bmatrix} \begin{bmatrix} \\ 312201311 \end{bmatrix}$.

[Hint: Recall the steps to calculate the determinant, given in Key Point 1.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Which of the following is the inverse matrix of $\begin{bmatrix} 323 \\ 101 \\ 211 \end{bmatrix} \begin{bmatrix} \\ 312201311 \end{bmatrix}$?

[Hint: If determinant is non-zero then the inverse of AA exists and is given by $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$
 $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$.]

☐

$$\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \\ 111-1-3-1-202 \end{bmatrix}$$

☐

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} \\ -1121-3011-2 \end{bmatrix}$$

☐

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} \\ -1111-3120-2 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \\ 1-1-1-13-1-202 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{2} \begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} -1111-3120-2 \end{bmatrix}$$

1 point

Choose the set of correct options.

Hint:

- $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$ $A^{-1} = \det(A) \text{adj}(A)$.
- $\det(A^{-1}) = \frac{1}{\det(A)} \det(A^{-1}) = \det(A) 1$

☐

If $A = A^{-1} A = A^{-1}$, then $\det(A) \det(A)$ must be 1.

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If $A = A^{-1} A = A^{-1}$, then AA must be the identity matrix.

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If $A = A^{-1} A = A^{-1}$, then $A^2 = IA^2 = I$.

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